Residual Certificates at the Static Reconstruction Step of Spacetime Penrose Rigidity

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Abstract

We analyze the equality manuscript *Equality and rigidity in the spacetime Penrose inequality*, keeping its original-data variation, static-base construction, conformal-double rigidity, and horizon recovery as distinct steps. On a fixed Euclidean spherical exterior we prove a sharp residual certificate for the minimal-boundary Robin problem occurring at the final conformal reconstruction. Its negative boundary coefficient is controlled by a sharp exterior trace inequality; exact spherical modes separate the mass coefficient from shape error and give certified omitted-mode bounds. A weighted source identity provides the additional mass control that the energy norm alone cannot supply. We give an explicit nonradial example with two scalar-flat conformal metrics and harmonic lapse but nonzero static tensor residual. These are local analytic results in a specified gauge, not a proof that a small Penrose deficit produces that gauge, a round horizon, or a Schwarzschild embedding. Classical harmonic analysis and weak-inverse arguments are credited explicitly. Contribution assessment: C3/5, the sharp horizon-residual and independent mass-channel certificates give a substantive fixed-gauge refinement while leaving global rigidity and near-equality compactness unproved.

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1 The Source Claim and Its Logical Boundaries

The assigned manuscript [1] asserts original-data rigidity, not merely rigidity of an auxiliary scalar-flat metric. We attribute that global assertion to the source; this report does not independently certify its entire proof. Put $n \geq 3$, $d = n-2$, $k = n-1$, and $\omega = |\mathbb{S}^{n-1}|$. The source uses

$$2\mu = R_g + (\operatorname{tr}_g K)^2 - |K|_g^2, \qquad J = \operatorname{div}_g(K - (\operatorname{tr}_g K)g), \qquad \mu \geq |J|_g.$$

The Laplacian is $\Delta = \operatorname{div}\nabla$. The boundary normal points from the horizon into the exterior, and $\theta_+ = H + \operatorname{tr}_S K$. No condition on the other null expansion is silently added. For the strong-decay branch the ADM normalization is

$$E = \frac{1}{2k\omega} \lim_{\rho\to\infty} \int_{S_\rho} (\partial_j g_{ij} - \partial_i g_{jj}) n_\delta^i \, \mathrm{d}A_\delta, \tag{1.1}$$

$$P_i = \frac{1}{k\omega} \lim_{\rho\to\infty} \int_{S_\rho} (K_{ij} - (\operatorname{tr}_g K) g_{ij}) n_\delta^j \, \mathrm{d}A_\delta, \qquad m = \sqrt{E^2 - |P|_\delta^2}. \tag{1.2}$$

The three-dimensional weak-decay modules instead write $16\pi\mu = R + \tau^2 - |K|^2$ and $8\pi J = \operatorname{div}(K - \tau g)$. Their densities are $1/(8\pi)$ times the preceding geometric densities; the ADM charges are unchanged.

A full enclosing cut is the entire compact boundary of the connected closed side containing the distant end. All its components and all portions coincident with the original boundary count. Its infimum $a(g)$ is computed in the original metric. The source imports

$$m \geq F_n(a(g)), \qquad F_n(a) = \frac{1}{2}(a/\omega)^{d/k}. \tag{1.3}$$

In its equality classes, the boundary is marginally outer trapped and outer area-minimizing, so $a(g) = A = |S|_g > 0$. This is not a definition equating every coordinate sphere's area with enclosing area.

Branch	End regularity	Additional distinctions
All $n \geq 3$	$g - \delta = O_6(r^{-q})$, $K = O_5(r^{-1-q})$, $d/2 < q < d$	Connected horizon, wholly-interior outermostness, complete closed exterior with one end and compact complement; $E > \lvert P\rvert$.
Weak $n = 3$	$O_2(r^{-q})/O_1(r^{-1-q})$, $q > 1/2$	Complete boundaryless ambient data with finitely many AF ends; partial-coincidence outermostness for finitely many horizon components. For connected S, wholly-interior outermostness suffices.
Weak $n = 4$	$O_2(r^{-q})/O_1(r^{-1-q})$, $q > 1$	Connected horizon and wholly-interior outermostness; no ambient extension assumption. Timelike ADM vector follows from the numerical input.

Smoothness, orientability, completeness with boundary, integrable constraint densities and finite ADM limits remain part of the hypotheses. The conclusions concern the chosen exterior, not the region behind the horizon. Equality permits nonzero K and an asymptotic boost. Disconnected-boundary strictness in the source is qualitative, without a uniform positive deficit depending only on component count.

1.1 Proof architecture and retained obligations

The root file and its nested inclusions give three equality branches. The cached inventory has 26 text files, including the bibliography and README. The functional route is as follows.

1. The setting and shared enclosure geometry define the full-cut infimum. Compactness and continuity of tangent-plane measures give

$$a'(g;h) = \min_{C\in\mathcal{T}(g)} \frac{1}{2}\int_{\partial^* C} \operatorname{tr}_{T\partial^* C} h \,\mathrm{d}A_g.$$

   This is a directional derivative over the full minimizing family.

2. Original-data variation uses the numerical inequality for nearby weakly trapped data, not only for nearby members of the rigidity class. Positive separation gives lapse, shift, and an area measure. Localization requires normal tests, outermostness, and, in high dimensions, the singular-sheet argument.

3. The homogeneous adjoint and its boundary laws construct a stationary field. Complementarity initially permits null stress. The twist identity has separate cutoffs at the interior null set, the horizon, and infinity; the source takes those limits in that order.

4. The common static base satisfies

$$\operatorname{Hess}_h \lambda = \lambda \operatorname{Ric}_h, \quad \Delta_h\lambda = 0, \quad R_h = 0, \quad \lambda|_S = 0, \quad \partial_{\nu_h}\lambda = \kappa > 0. \tag{1.4}$$

   It is totally geodesic at S. The strong branch excludes additional null ends by its circle completion and splitting argument. The weak branches retain possible ends until their respective classification steps.

5. Harmonic-coordinate asymptotics justify the regularity of $h_\pm = ((1\pm\lambda)/2)^{4/d}h$, the zero mass of the plus end, and compactification of the minus end. Rough-double positive-mass and rigidity arguments are additional global obligations, not consequences of finite residual calculations.

6. Only after the double is Euclidean does the source locate a round seam, solve an exterior harmonic problem, and recover the original time graph. Its logarithmic time primitive requires horizon-regular coordinates; smoothness in static time is not the claimed conclusion.

The three- and four-dimensional modules redo variation, static asymptotics and classification with weaker decay. The separate local spinor section treats a two-component obstruction. Their omitted proofs are explicitly listed in the read ledger, not counted as checked.

1.2 Where this report enters

In the strong branch, `static-main.tex`, lines 1151–1207, gives

$$\delta = \left(\frac{1+\lambda}{2}\right)^{4/d} h, \quad \Psi = \frac{2}{1+\lambda}, \quad \Delta_\delta\Psi = 0, \quad \Psi|_{r=R} = 2, \quad \Psi \to 1. \tag{1.5}$$

The unique solution is

$$\Psi_0 = 1 + (R/r)^d, \quad h_0 = \Psi_0^{4/d}\delta, \quad \lambda_0 = 2/\Psi_0 - 1, \quad M_0 = 2R^d. \tag{1.6}$$

The areal horizon radius is $2^{2/d}R$, not R. The weak branches obtain the same problem after their separate global steps. Our results start with a fixed center, $R > 0$, and this Euclidean exterior. They do not supply those global steps.

2 Scalar Tests Versus the Full Static Equation

The scalar part of a static reconstruction can be tested exactly, but it is not the whole system.

Proposition 2.1 (A scalar pair and its missing tensor)**.** *Let $\Psi > 0$ be C^3 on a Euclidean open set and put $h = \Psi^{4/d}\delta$, $\lambda = 2/\Psi - 1$. Then*

$$R_h = -\frac{4(n-1)}{d}\Psi^{-(n+2)/d}\Delta\Psi, \qquad \Delta_h\lambda = -2\Psi^{-2n/d}\Delta\Psi. \tag{2.1}$$

If $\Delta\Psi = 0$, writing $v = \Psi - 1$ gives

$$\operatorname{Hess}_h \lambda - \lambda \operatorname{Ric}_h = \frac{2}{\Psi^2}\left[-vD^2v + \frac{n}{d}\,dv \otimes dv - \frac{1}{d}|dv|_\delta^2\delta\right]. \tag{2.2}$$

Where $v > 0$, both $h_+ = \delta$ and $h_- = v^{4/d}\delta$ are scalar flat; this does not force (2.2) to vanish.

Proof. For $h = e^{2f}\delta$, $f = (2/d)\log\Psi$, the conformal scalar formula gives the first identity. Also $\Delta_h a = \Psi^{-4/d}(\Delta a + 2\Psi^{-1}\nabla\Psi\cdot\nabla a)$. For $a = 2/\Psi - 1$ the squared-gradient terms cancel. When $\Delta\Psi = 0$, direct differentiation gives

$$\operatorname{Ric}_h = -2\Psi^{-1}D^2\Psi + \frac{2n}{d}\Psi^{-2}d\Psi^2 - \frac{2}{d}\Psi^{-2}|d\Psi|^2\delta,$$
$$\operatorname{Hess}_h \lambda = -2\Psi^{-2}D^2\Psi + \frac{4n}{d}\Psi^{-3}d\Psi^2 - \frac{4}{d}\Psi^{-3}|d\Psi|^2\delta.$$

Subtract $(2/\Psi - 1)\operatorname{Ric}_h$ to obtain the tensor formula. The identities for $h_\pm$ follow by substitution; their scalar curvatures vanish because 1 and v are harmonic. □

Example 2.2 (Exact scalar agreement without staticity)*.* In dimension three take

$$v(x) = \frac{1}{r} + t\frac{x_1}{r^3}, \qquad 0 < |t| < \frac{1}{2}, \qquad r > 2.$$

Then $0 < v < 1$, so $0 < \lambda < 1$, and every scalar test in Proposition 2.1 vanishes. At $x = (0, r, 0)$, however,

$$\operatorname{Hess}_h \lambda - \lambda \operatorname{Ric}_h = \frac{2t^2}{(1 + 1/r)^2 r^6}\ \operatorname{diag}(2, -1, -1) \neq 0. \tag{2.3}$$

Indeed $dv = (t/r^3, -1/r^2, 0)$, $D^2v = r^{-3}\operatorname{diag}(-1, 2, -1)$ apart from $v_{12} = v_{21} = -3t/r^4$. Substitution proves the formula. This exterior example is not asserted to have the source's horizon conditions. It disproves only the promotion from the scalar equations to staticity. It does not contradict the source, which retains the tensor equation.

3 The Horizon Robin Problem in the Correct Energy Space

Let $\Omega_R = \{x \in \mathbb{R}^n : r > R\}$, $n \geq 3$, with Euclidean volume and surface measures. Define V as the completion, in $\|\nabla u\|_{L^2(\Omega_R)}$, of smooth functions on $\overline{\Omega_R}$ compactly supported at infinity. No zero trace is imposed at S_R. This homogeneous energy space embeds locally in H^1 and in $L^{2n/d}$; a constant at infinity is excluded. An annular extension, the trace bound below, and the Euclidean Sobolev inequality give these standard embeddings.

For a positive smooth Ψ, conformal mean curvature with normal $+\partial_r$ is

$$H_{\Psi^{4/d}\delta} = \Psi^{-2/d}\left(\frac{k}{R} + \frac{2k}{d}\frac{\partial_r\Psi}{\Psi}\right) = \frac{2k}{d}\Psi^{-1-2/d}q, \qquad q = \partial_r\Psi + \frac{d}{2R}\Psi. \tag{3.1}$$

Thus $q = 0$ is minimality on the already fixed sphere. It is not Dirichlet data, and its coefficient in the weak exterior form is negative because the integration normal there is $-\partial_r$. For Ψ_0 in (1.6), $q = 0$.

Lemma 3.1 (Sharp exterior coercivity)**.** *For every $u \in V$,*

$$\frac{d}{R}\int_{S_R} |u|^2 \,\mathrm{d}A \le \int_{\Omega_R} |\nabla u|^2 \,\mathrm{d}x, \qquad \frac{1}{2}\|\nabla u\|_2^2 \le a_0(u,u) \le \|\nabla u\|_2^2, \tag{3.2}$$

where

$$a_0(u,z) = \int_{\Omega_R} \nabla u \cdot \nabla z \,\mathrm{d}x - \frac{d}{2R}\int_{S_R} uz \,\mathrm{d}A.$$

The constants d/R and $1/2$ are sharp.

Proof. For each angle, $u(R,\theta) = -\int_R^\infty \partial_r u \,\mathrm{d}r$. Cauchy–Schwarz with weights r^{n-1} and r^{1-n} gives

$$|u(R,\theta)|^2 \le \frac{R^{-d}}{d}\int_R^\infty |\partial_r u|^2 r^{n-1} \,\mathrm{d}r.$$

Integrate angles and multiply by R^{n-1}; density gives the result on V. The form bounds follow. The radial function $(R/r)^d$ belongs to V, by cutoff approximation at infinity, and attains equality in both lower bounds. □

Theorem 3.2 (Full weak residual, including the horizon)**.** *For every continuous real linear functional $L \in V'$ there is a unique $w \in V$ with $a_0(w,z) = L(z)$ for every $z \in V$. For any candidate $\widehat{w} \in V$, put $\mathcal{R}(z) = L(z) - a_0(\widehat{w}, z)$. Then*

$$\|\nabla(w - \widehat{w})\|_2 \le 2\|\mathcal{R}\|_{V'}. \tag{3.3}$$

The factor 2, with these norms, cannot be decreased. For classical data $\Delta w = f$ and $\partial_r w + dw/(2R) = q$, the load is

$$L(z) = -\int_{\Omega_R} fz \,\mathrm{d}x - \int_{S_R} qz \,\mathrm{d}A. \tag{3.4}$$

The pairings must extend to V; for example $f \in L^{2n/(n+2)}$ and $q \in L^2(S_R)$ suffice.

Proof. The bounded coercive form gives existence and uniqueness by the classical weak-inverse argument. Test the error equation with the error. For the radial extremizer, harmonic integration by parts gives $a_0(w,z) = \frac{1}{2}\int \nabla w \cdot \nabla z$, proving sharpness. For general data use $\partial_{\nu_{\Omega_R}} w = -q + dw/(2R)$ to obtain (3.4). Boundary residuals cannot be omitted. □

This is not a new Lax–Milgram theorem. The original Ram weak-operator proof [2] warns that negative Robin coefficients need an independent inverse bound. Here (3.2) supplies it on the specified noncompact domain; a finite matrix's invertibility would not.

4 Exact Modes, Shape Error, and Certified Truncation

Use the probability measure $\mathrm{d}\sigma/\omega$ on the unit sphere. Let $Y_{\ell j}$ be a real orthonormal basis of degree-ℓ spherical harmonics, with $Y_{01} = 1$. Set

$$N_\ell = \binom{\ell+n-1}{n-1} - \binom{\ell+n-3}{n-1},$$

where a binomial coefficient with top smaller than bottom is zero. The Euclidean angular eigenvalue is $\ell(\ell+d)$. Pull a boundary residual q back by $x = R\theta$ and define $t(\theta) = -Rq(R\theta)$, with coefficients $t_{\ell j}$. The dimensionless variable t, rather than an unscaled coordinate norm, will be used in the certificate.

Theorem 4.1 (Sharp harmonic horizon certificate)**.** *For $q \in L^2(S_R)$, the unique V-solution of $\Delta w = 0$, $\partial_r w + dw/(2R) = q$, is*

$$w(r,\theta) = \sum_{\ell,j} b_{\ell j}(R/r)^{\ell+d} Y_{\ell j}(\theta), \qquad b_{\ell j} = \frac{t_{\ell j}}{\ell + d/2}. \tag{4.1}$$

The series converges in energy and smoothly on every exterior buffer. Its exact Dirichlet energy and Robin-form energy are

$$\frac{\|\nabla w\|_2^2}{\omega R^d} = \sum_{\ell,j} \frac{\ell+d}{(\ell+d/2)^2}|t_{\ell j}|^2, \tag{4.2}$$

$$\frac{a_0(w,w)}{\omega R^d} = \sum_{\ell,j} \frac{|t_{\ell j}|^2}{\ell+d/2}. \tag{4.3}$$

In particular,

$$\|\nabla w\|_2^2 \le \frac{4R}{d}\|q\|_{L^2(S_R)}^2. \tag{4.4}$$

This constant is sharp on the monopole. If the boundary mean of q vanishes, it improves to

$$\|\nabla w\|_2^2 \le \frac{4R(d+1)}{(d+2)^2}\|q\|_{L^2(S_R)}^2,$$

sharply on degree one.

Proof. Separation of variables in the Euclidean Laplacian gives the decaying radial solution $r^{-\ell-d}$; the growing solution is excluded by membership in V. At $r = R$ the Robin derivative is $-(\ell+d/2)b_{\ell j}/R$. Orthogonality and integration by parts give

$$\int_{\Omega_R} \left|\nabla\left[b(R/r)^{\ell+d}Y_{\ell j}\right]\right|^2 \mathrm{d}x = \omega R^d(\ell+d)|b|^2.$$

For a direct check, the radial and angular contributions have numerator $(\ell+d)^2 + \ell(\ell+d)$ and radial integral $R^{-2\ell-d}/(2\ell+d)$; their product reduces to $\ell+d$. Subtract the boundary term to obtain (4.3). These sums are finite for $t \in L^2$; the partial sums converge in V and solve the weak equation in the limit. Uniqueness follows from Theorem 3.2. Interior harmonic estimates give smooth convergence off the boundary. Finally $F(x) = (x+d)/(x+d/2)^2$ has derivative $-(x+3d/2)/(x+d/2)^3 < 0$. Its maxima on $\ell \ge 0$ and $\ell \ge 1$ are $4/d$ and $4(d+1)/(d+2)^2$, respectively. Use $\|q\|_{L^2(S_R)}^2 = \omega R^{d-1}\sum|t_{\ell j}|^2$. □

Corollary 4.2 (A finite certificate with a stated complement)**.** *Suppose coefficients $\widehat{t}_{\ell j}$ are supplied for $\ell \le J$, and certified errors $e_{\ell j}$ and T satisfy*

$$|t_{\ell j} - \widehat{t}_{\ell j}| \le e_{\ell j}, \qquad \sum_{\ell > J, j} |t_{\ell j}|^2 \le T^2.$$

Let $\widehat{w}_J$ be (4.1) *with those supplied coefficients and zero higher coefficients. Then*

$$\frac{\|\nabla(w - \widehat{w}_J)\|_2^2}{\omega R^d} \le \sum_{\ell \le J, j} F(\ell) e_{\ell j}^2 + F(J+1) T^2. \tag{4.5}$$

The complementary constant $F(J+1)$ is optimal for this information. For $r \ge sR$, $s > 1$, the unmeasured part alone satisfies

$$|w_{>J}(r,\theta)| \le T \sqrt{\mathcal{K}_{n,J}(s)}, \tag{4.6}$$

$$\mathcal{K}_{n,J}(s) = \sum_{\ell > J} \frac{N_\ell s^{-2(\ell+d)}}{(\ell + d/2)^2} \le \frac{s^{-2d}}{(J+1+d/2)^2} \left[\frac{1+z}{(1-z)^{n-1}} - \sum_{\ell=0}^{J} N_\ell z^\ell \right], \quad z = s^{-2}. \tag{4.7}$$

Proof. Use the exact energy identity on the difference and monotonicity of F. Concentrating all tail energy in one degree-$J+1$ harmonic proves optimality. For pointwise control use Cauchy–Schwarz and the addition identity $\sum_j Y_{\ell j}(\theta)^2 = N_\ell$, valid for our probability normalization. Summing the dimension formula gives $\sum_{\ell \ge 0} N_\ell z^\ell = (1+z)/(1-z)^{n-1}$. All terms are nonnegative, so the denominator bound proves (4.7). No unproved interchange at $s = 1$ is used. □

The tail premise is essential. Choosing a single harmonic of degree $J+1$ makes every measured coefficient zero while leaving positive energy. Finite moments alone do not supply T. In a nonorthogonal finite frame, the coefficients in these formulas must be converted using the actual spherical Gram matrix; the omitted orthogonal component remains. This is precisely the physical-residual distinction proved in the original metric chapter [3]. We do not credit that classical distinction itself as a new result.

4.1 Mass channel and a worked mode

For $\Psi = \Psi_0 + w > 0$, the harmonic expansion has

$$\Psi = 1 + a_\infty r^{-d} + O(r^{-d-1}), \qquad a_\infty = R^d(1 + b_{01}).$$

Substitution into the ADM integral for $\Psi^{4/d}\delta$ gives

$$M_{\mathrm{ADM}} = 2a_\infty, \qquad M_{\mathrm{ADM}} - M_0 = \frac{4R^d}{d} t_{01} = -\frac{4R^{d+1}}{d}\, \mathbb{E}q(R\theta). \tag{4.8}$$

Indeed the leading metric coefficient is $(4a_\infty/d) r^{-d}\delta_{ij}$; its ADM integrand is minus k times the radial derivative of that scalar coefficient. This gives $2a_\infty$ with the normalization of Section 1. This is the Riemannian mass of the candidate base. Identification with the invariant ADM mass of original nonzero-K data is an additional source conclusion, not a consequence of this computation.

In $n = 3, R = 1$, take $t = t_0 + t_2 Y_{2j}$. Then

$$b_{01} = 2t_0, \quad b_{2j} = \frac{2}{5} t_2, \quad \frac{\|\nabla w\|_2^2}{4\pi} = 4t_0^2 + \frac{12}{25} t_2^2, \quad M_{\mathrm{ADM}} = 2 + 4t_0.$$

With $t_0 = 1/100, t_2 = 1/50$ the normalized energy is $37/62500$ and the mass is $51/25$. These are exact algebraic values for a manufactured harmonic fixture, not measurements of an unknown equality geometry. A sufficiently small finite mode is positive after adding Ψ_0; the addition theorem supplies an explicit sup-norm bound if required.

5 Volume Defects and the Independent Mass Functional

Theorem 5.1 (The weighted source and horizon mass identity)**.** *Let w be C^2 on $\overline{\Omega_R}$, tending to zero, with $f = \Delta w$, $q = \partial_r w + dw/(2R)$, and*

$$w = a_w r^{-d} + O_1(r^{-d-\eta}), \qquad \eta > 0, \qquad \int_{\Omega_R} \Psi_0 |f| \, \mathrm{d}x < \infty.$$

Then

$$a_w = -\frac{1}{d\omega} \left(\int_{\Omega_R} \Psi_0 f \, \mathrm{d}x + 2 \int_{S_R} q \, \mathrm{d}A \right). \tag{5.1}$$

If $\Psi = \Psi_0 + w > 0$, its ADM mass obeys

$$|M_{\mathrm{ADM}} - 2R^d| \le \frac{2}{d\omega} \left(\int_{\Omega_R} \Psi_0 |f| \, \mathrm{d}x + 2 \int_{S_R} |q| \, \mathrm{d}A \right). \tag{5.2}$$

For an integral computed only out to L, the additional mass error is at most $2(d\omega)^{-1} \int_{r>L} \Psi_0 |f| \, \mathrm{d}x$.

Proof. Apply the second Green identity to Ψ_0, w on $R < r < L$. At the inner boundary, where the integration normal is $-\partial_r$,

$$\Psi_0 \partial_\nu w - w \partial_\nu \Psi_0 = -2\partial_r w - \frac{d}{R} w = -2q.$$

At infinity the flux tends to $-d\omega a_w$; the term involving $w\partial_r \Psi_0$ tends to zero. Since $\Delta \Psi_0 = 0$, the integral of $\Psi_0 f$ equals this outer flux minus $2 \int q$. This proves the signed identity. The absolute estimate and the truncated version follow. The ADM calculation is the one above. □

This functional tests f against Ψ_0, which tends to one and does *not* belong to V. It is therefore extra information beyond $\|f\|_{V'}$, not a hidden consequence of coercivity. The source's compact-source inverse and flux formula (`rough-mass.tex`, lines 72–188) likewise retain source integrals. Our exterior formula adds the horizon's signed contribution and its nonconstant harmonic weight.

Proposition 5.2 (Small energy can miss a fixed mass)**.** *There are smooth $w_L \in V$ that vanish in $r < L$, have $q = 0$ on S_R, and satisfy*

$$\|\nabla w_L\|_2 \longrightarrow 0, \qquad \|a_0(w_L, \cdot)\|_{V'} \longrightarrow 0, \qquad a_{w_L} = a > 0$$

for any fixed a and $L \to \infty$. Thus no uniform estimate of the mass coefficient by this energy residual is possible without a tail condition.

Proof. Fix a smooth cutoff χ, zero on $(-\infty, 1]$, one on $[2, \infty)$, and between zero and one. Let $w_L = a r^{-d} \chi(r/L)$, $L > 2R$. Scaling $r = Ls$ gives $\|\nabla w_L\|_2^2 = C_\chi a^2 L^{-d}$. The constant is finite, including the harmonic tail $r > 2L$. The trace vanishes, so $\|a_0(w_L, \cdot)\|_{V'} \le \|\nabla w_L\|_2$. The coefficient at infinity is exactly a, and (5.1) gives $\int \Psi_0 \Delta w_L = -d\omega a$. The missing weighted integral therefore does not tend to zero. □

This example is not asserted to satisfy the dominant energy condition: its transition source can have both signs. It identifies an analytic failure of a proposed residual-to-mass implication, not a counterexample to the Penrose inequality. Local convergence on every fixed compact set also misses its mass.

6 A Controlled Metric and Boundary Perturbation

An approximate Euclidean gauge must not be treated as exact. Suppose a symmetric coefficient field $A(x)$ and a real boundary coefficient b satisfy

$$\|A - I\|_{L^\infty(\mathrm{op})} \le \varepsilon, \qquad \frac{R}{d}\|b\|_{L^\infty(S_R)} \le \zeta, \qquad \theta := \varepsilon + \zeta < \frac{1}{2}.$$

Use the fixed measures $\mathrm{d}x$, $\mathrm{d}A$, and define

$$a_{A,b}(u,z) = \int A\nabla u \cdot \nabla z - \int_{S_R} \left(\frac{d}{2R} + b\right) uz.$$

Proposition 6.1 (A validated perturbation radius)**.** *The form is coercive on V, with constant $1/2 - \theta$. For an exact weak solution w of load L and any candidate $\widehat{w} \in V$,*

$$\|\nabla(w - \widehat{w})\|_2 \le \frac{\|L - a_{A,b}(\widehat{w}, \cdot)\|_{V'}}{1/2 - \theta}. \tag{6.1}$$

If w_0 solves $a_0(w_0, \cdot) = L_0$, then

$$\|\nabla(w - w_0)\|_2 \le \frac{\|L - L_0\|_{V'} + \theta\|\nabla w_0\|_2}{1/2 - \theta}.$$

Proof. The coefficient error is at most $\varepsilon\|\nabla u\|_2\|\nabla z\|_2$. The trace inequality bounds the boundary error by the same product times ζ. Subtract this total from (3.2) and apply the weak-inverse proof. For the last estimate the residual of w_0 is $L - L_0 + (a_0 - a_{A,b})(w_0, \cdot)$. □

This is a sufficient, not optimal, perturbation radius. On a pulled-back Riemannian metric γ, the coefficient is $A = \sqrt{\det\gamma}\,\gamma^{-1}$, not just γ^{-1}. The boundary Jacobian and physical Robin coefficient must both be absorbed into b; any boundary-shape change needs an actual diffeomorphism and quantitative coefficient bounds. No such global map is constructed here. The mass formula of Theorem 5.1 belongs to the Euclidean conformal model and is not silently transferred to $A \neq I$. The original Ram perturbation proof [4] motivates keeping operator and load errors distinct; the exterior constants here come from the explicit trace calculation.

7 What the Certificates Do Not Promote

7.1 Boundary area is not enclosing area

For a harmonic candidate write $b = w|_{S_R}$. Its coordinate sphere has area

$$A_{\mathrm{sph}} = \omega R^k\, \mathbb{E}(2 + b)^p, \qquad p = \frac{2k}{d}. \tag{7.1}$$

If $\|b\|_\infty \le \eta < 2$, this gives the rigorous interval $\omega R^k(2 - \eta)^p \le A_{\mathrm{sph}} \le \omega R^k(2 + \eta)^p$. Such a sup bound requires regularity or a separate boundary tail certificate; (4.7) only applies on an interior buffer. Nothing here asserts that this sphere realizes $a(h)$.

For an explicit warning, in $n = 3, R = 1$ extend $\Psi = 1 + r^{-1} + tx_1/r^3$ to all $r \ge 1$, with $0 < |t| < 1/2$. It is positive and harmonic. Its mass is exactly 2, while

$$A_{\mathrm{sph}} = 4\pi\left(16 + 8t^2 + \frac{t^4}{5}\right), \qquad F_3(A_{\mathrm{sph}}) = \frac{1}{2}\sqrt{16 + 8t^2 + t^4/5} > 2.$$

Here $\mathbb{E}\theta_1^2 = 1/3$, $\mathbb{E}\theta_1^4 = 1/5$. The explanation is explicit: $q = -3t\theta_1/2$, so the boundary is not a minimal surface, and with $K = 0$ it is not a MOTS. There is no contradiction to (1.3). Even a small, exactly known Robin error is not the exact admissibility hypothesis.

7.2 The envelope has no automatic quadratic remainder

The source correctly claims a directional derivative, not a uniform second-order expansion of the minimum. As an elementary abstract example, take $f_s(t) = a + s^4 - ts$ for $0 \le s \le 1$. Every competitor is affine in t; the unique minimizer at $t = 0$ is $s = 0$. Nevertheless, for $0 < t < 4$,

$$\min_s f_s(t) = a - 3\,4^{-4/3} t^{4/3}.$$

The minimizer is $s = (t/4)^{1/3}$, obtained by differentiation. Thus a uniform competitor Taylor remainder and the directional derivative do not imply an $O(t^2)$ envelope remainder. This is a scalar variational obstruction, not a constructed geometric counterexample to the source's enclosure theorem. Quantitative near-equality separation would need control of near-minimizers and of the feasible perturbation radius.

7.3 Global gaps and gauges

The energy norm is a fixed-gauge norm, not a distance between unparametrized geometries. Our center and radius are prescribed; we have not estimated them from approximately umbilic data. We have not bounded the source's lapse–shift multiplier or its area measure in terms of $m - F_n(a)$, controlled the null-set cutoff uniformly over nearly equal data, or proved compactness of their doubles. Small scalar curvature does not prove Ricci flatness, and small Ricci residual does not supply completeness or the equality case of a positive-mass theorem.

The source's harmonic-coordinate inverse is obtained sufficiently far out in a *fixed* end. Its compact-source inverse is uniform for small perturbations of a *fixed* compact region with the end unchanged (`rough-mass.tex`, lines 94–96 and 184–187). Neither statement supplies uniform constants over arbitrary near-equality families, degenerating collars or moving sources. Proposition 5.2 shows exactly why a tail cannot be replaced by a fixed collection of interior measurements. Finally, the original slice is recovered from $A_s = X^\flat/(u^2 - |X|^2)$, its global primitive, and its horizon asymptotics. None of our estimates constructs that primitive or the spacelike embedding.

8 Comparison, Advantages, and Further Work

8.1 What is inherited and what is added

The source already proves, within its asserted global argument, the static end expansion, conformal compactification, round-seam identification and uniqueness of the exterior harmonic Dirichlet problem. Those are not new results of this report. The additions are the explicitly sharp minimal-boundary residual inverse, its physical mode weights and complement budget, the signed weighted-source mass channel, and the exact examples separating these from invalid scalar-to-static and energy-to-mass promotions.

The private original proofs provide methodological support with strict scope. TR8 [2] supplies the weak-domain, boundary-lifting and Green-inverse distinction. Its bounded-domain theorem is not directly applied to Ω_R; Lemma 3.1 establishes the needed exterior coercivity independently. The metric chapter [3] proves physical Gram identities and the omitted residual formula. TR1 [4] separates load and operator perturbations and retains boundary terms in weak integration by parts. None proves the spacetime Penrose equality theorem, and none turns finite jets into a global elliptic inverse. M518's enclosing-cut transfer and graph-penalty results concern a different manuscript and are not credited again here.

The classical ingredients are spherical harmonics, the exterior Dirichlet-to-Neumann map, Lax–Milgram, Green's identity, and conformal curvature formulas. We claim a source-specific assembly and quantitative refinement, not historical priority for these ingredients. Applicability remains L3. The C3 assessment concerns the proved local certificate and its limitations, not validation of the source's all-dimensional theorem.

8.2 Public context checked on October 9, 2026

The public primary records were checked at abstract/version level; their full proofs were not independently audited here. Gibbons–Ida–Shiromizu [5] address asymptotically flat static vacuum uniqueness, a classical context for the source's conformal construction, not a theorem giving a static base from arbitrary nearly equal initial data. Lee–Sormani [6] study rotationally symmetric near-equality and explicitly allow a Schwarzschild space with an appended cylinder. Dong's Riemannian stability theorem [7] uses pointed measured Gromov–Hausdorff convergence modulo negligible domains and boundary-area perturbations. These different conclusions emphasize why a topology and admissible degenerations must be specified.

Allen–Bryden–Kazaras–Khuri [8] give a three-dimensional spacetime bound with a suboptimal universal constant and enclosing area. Dong's July 2026 preprint [9] has a connected outermost past apparent horizon and a 2-convexity assumption; its v2 record explicitly notes the connectedness addition. Hirsch–Huang [10] discuss mass-minimizing initial data and strongly stationary developments. The source's full-area variational problem is not identified with fixed Bartnik boundary data. These records are evidence of differing theorem scopes, not independent verification of the assigned manuscript.

8.3 Benefits, costs, and feasible extensions

The advantages are explicit signs, scale dependence, sharp monopole and mean-zero constants, and a finite stopping rule with an honest unmeasured remainder. The cost is a fixed spherical Euclidean gauge; the mass estimate additionally requires a signed integral or a weighted absolute tail. There is no measured computational speedup.

A useful next step would be a validated boundary diffeomorphism from a quantitatively nearly umbilic seam, followed by the pulled-back coefficient bounds of Proposition 6.1. This requires injectivity and geometric regularity estimates, not merely small entries of a sampled shape-operator matrix. Another is a joint certificate for (2.2), the horizon data and the asymptotic source moment. A global near-equality theorem would still need uniform feasibility, enclosure, null-set, compactification, and positive-mass stability estimates. This report supplies none of those missing uniform geometric statements.

9 Verification and Read Scope

The public subtree contains 13,109 lines in 26 text files. Fresh literal bounded displays covered 3,718 lines in 15 of those files, including the complete 1,325-line common static module, the full introduction and setting, the weak-branch theorem statements, and the specified enclosure, variation, mass-inverse and classification passages. Architecture searches are recorded separately and are not counted as proof reading. The omitted 9,391 public lines, including most dimension-specific variation and adjoint proofs, have exact complements in the read ledger. No whole-manuscript or whole-corpus audit is claimed.

Fresh original private reading covered 430 lines: TR8, lines 81–195; the physical-metric chapter, lines 1–210; and TR1's `stability.tex`, lines 1–105. Pre-read hashes were persisted before those displays. Post-read hashes and captured-tree comparisons check byte identity only, not mathematical validity. Confidential material remained local.

The owned executable checks use exact rational arithmetic, including second-order automatic differentiation, for mode integration, boundary signs, sharp constants, finite tail bounds, source flux, the dipole tensor, and the area example. Counts and the exact execution environment are in the accompanying check receipt. The run passed 2,433 parameter cases and 8,621 counted mathematical assertions, with zero floating-point fixtures and zero parent cases. They are algebraic fixtures, not numerical PDE solutions or global rigidity certification. Parent tests are not counted as author tests. The local proofs above, rather than the finite fixture count, establish the general statements.

10 Conclusion

The final reconstruction step admits a sharp, physically normalized residual analysis once its geometry and gauge have genuinely been established. Its energy, mass and tensor channels answer different questions. Preserving those distinctions yields a useful certificate without promoting approximate scalar agreement into spacetime rigidity.

References

[1] OpenAI. *Equality and rigidity in the spacetime Penrose inequality*, September 27, 2026. Assigned cached source, pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Public source directory. Captured-tree provenance is separate from proof verification.

[2] M. Subbarao. Original private *TR8-lin-non-lin*, in `RamNonlin-IE-PDE-astra`; `TR8-lin-non-lin.tex`, lines 81–195. Weak elliptic operators, boundary domains and Green equivalence.

[3] M. Subbarao. Original private *RamCalBooks*, volume 21, `ch01d-physical-metrics.tex`, lines 1–210. Physical Gram norms, full residuals and finite-measurement limitations.

[4] M. Subbarao. Original private *TR1-lin-nonlin*, in `RamNonlin-IE-PDE-astra`; `stability.tex`, lines 1–105. Perturbation and weak boundary identities.

[5] G. W. Gibbons, D. Ida and T. Shiromizu. *Uniqueness and non-uniqueness of static vacuum black holes in higher dimensions*. Prog. Theor. Phys. Suppl. 148 (2003), 284–290. arXiv:gr-qc/0203004v2.

[6] D. A. Lee and C. Sormani. *Near-equality of the Penrose Inequality for rotationally symmetric Riemannian manifolds*. Ann. Henri Poincare 13 (2012), 1537–1556. arXiv:1109.2165.

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[8] B. Allen, E. Bryden, D. Kazaras and M. Khuri. *Proof of the Spacetime Penrose Inequality With Suboptimal Constant in the Asymptotically Flat and Asymptotically Hyperboloidal Regimes*. arXiv:2504.10641v1 (2025).

[9] C. Dong. *The Penrose conjecture for initial data sets satisfying a 2-convexity condition.* arXiv:2607.20741v2, July 30, 2026.

[10] S. Hirsch and L.-H. Huang. *Monotonicity of Causal Killing Vectors and Geometry of ADM Mass Minimizers.* arXiv:2510.10306v1 (2025).